### 1. Competency #2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

> *"Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem"*

- Highly relevant: This is the core skill assessed.
- All four statements require evaluating their plausibility using both data and real-world logic:
  - Statement 1: Requires reading values from the chart for 2000 and checking if they exceed 0.6 (direct data verification).
  - Statement 2: Interprets "average chicks < 0.8" as "fewer than 80% of pairs successfully raised a chick" — which is valid only if each pair raises at most one chick, a biologically reasonable assumption supported by prior context.
  - Statement 3 (key): The chart ends in 2008, and values—though declining—are still well above zero (e.g., Rockhopper ≈ 0.4 in 2008). Claiming "extinction by 2015" is a gross over-extrapolation. Students must recognize this as unfounded, demonstrating sensitivity to the limits of inference.
  - Statement 4: Requires verifying whether the Magellanic bars show a consistent decline from 2001 to 2004 (data observation).
- Especially for Statement 3, the task directly assesses whether students can reject an extreme conclusion that appears trend-based but lacks empirical support, perfectly aligning with Competency #2.

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### 2. Competency #3: Identify the limits of the model used to solve a problem

> *"Identify the limits of the model used to solve a problem"*

- Highly relevant: The flaw in Statement 3 stems precisely from ignoring the limitations of extrapolating limited data.
- The chart covers only nine years; predicting extinction seven years beyond that ignores critical ecological factors (habitat loss, food availability, climate change, etc.).
- Students must realize that short-term trends ≠ long-term fate, especially when values remain significantly above zero.
- Correctly labeling Statement 3 as False demonstrates understanding that "limited observational data does not justify deterministic predictions about extinction" —a direct manifestation of Competency #3.

### 3. Competency #9: Understand the relationship between the problem context and the mathematical representation, and use this to interpret the real-world meaning, feasibility, and limitations of the solution

> *"Understand the relationship between the context of the problem and representation of the mathematical solution... gauge the feasibility and possible limitations of the solution"*

- Strongly relevant: Students must connect the bar chart (mathematical representation) with penguin reproductive ecology (real-world context):
  - Recognize that "average chicks per pair" reflects breeding success;
  - Understand that even declining values (e.g., 0.4) imply some reproductive success—not population collapse;
  - Know that a value like 0.4 means roughly 40% of pairs succeeded (assuming max one chick per pair), not total failure.
- This bidirectional mapping between data and biological meaning is central to Competency #9.

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### 4. Competency #1: Interpret a mathematical result back into the real-world context

> *"Interpret a mathematical result back into the real world context"*

- Moderately relevant: For example, reading "average = 0.75 in 2006" must be understood as "most pairs successfully raised a chick," not just as a number.
- Statement 2 particularly relies on this: translating "average < 0.8" into "fewer than 80% of pairs had a chick" requires interpreting a statistical measure as a proportion of successful breeding events.

### 5. Competency #5: Interpret mathematical outcomes in a variety of formats and compare or evaluate multiple representations in relation to a situation

> *"Interpret mathematical outcomes in a variety of formats... compare or evaluate two or more representations in relation to a situation"*

- Moderately relevant: Although only one format (a bar chart) is used, the task involves:
  - Cross-species comparison (Gentoo vs. Rockhopper vs. Magellanic);
  - Temporal comparison (2001 vs. 2004);
  - Threshold comparison (>0.6? <0.8?).
- This constitutes multi-dimensional evaluation within a single visualization, fitting a broad interpretation of Competency #5.

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### 6. Competency #6: Use real-world knowledge to judge how mathematical results should be adjusted or applied

> *"Use knowledge of how the real world impacts the outcomes... to make contextual judgments about how the results should be adjusted or applied"*

- Weak but present: Rejecting Statement 3 implicitly draws on real-world understanding:
  - Species extinction is a complex process influenced by many factors beyond annual chick production;
  - Students aware of population resilience may more confidently judge the statement as false.
- While the task doesn't require actively "adjusting" the model, rejecting an unreasonable application itself reflects contextual judgment.

### 1. 第2项：判断数学结果或结论在给定问题情境下是否合理

> *"Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem"*

- 高度相关：这是本题最核心的能力。
- 所有四个陈述都需结合数据趋势与现实常识判断其合理性：
  - 陈述1：需从图中读取2000年各物种值并判断是否 >0.6（纯数据验证）；
  - 陈述2：将“平均雏鸟数 < 0.8”等价于“<80% 的配对成功育雏”——这仅在每对最多育1雏的前提下成立，而题干未明确说明，但生物学常识支持此假设，故可接受；
  - 陈述3（关键）：图表仅到2008年，且数值虽下降但仍显著高于0（如 Rockhopper 在2008年约0.4），外推至“2015年灭绝”是过度推断。学生需识别这种不合理外推，体现对模型局限性的敏感；
  - 陈述4：需观察 Magellanic 柱状图在2001–2004是否单调下降（数据验证）。
- 尤其是陈述3，直接考察学生是否能拒绝看似“有趋势”但实际无依据的极端结论，完美契合第2项。

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### 2. 第3项：识别所用模型的局限性

> *"Identify the limits of the model used to solve a problem"*

- 高度相关：陈述3的错误根源正是忽视模型（即数据外推）的局限性。
- 图表仅为9年观测数据，无法预测7年后的种群存续（灭绝涉及更多因素：栖息地、食物、气候等）；
- 学生需意识到：线性或趋势外推在生态预测中风险极高，尤其当数据未趋近于零时。
- 正确判断“False”即表明学生理解“有限数据 ≠ 长期命运”，直接体现第3项。

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### 3. 第9项：理解问题情境与数学解表示之间的关系，并据此解释解的现实含义及可行性/局限性

> *"Understand the relationship between the context of the problem and representation of the mathematical solution... gauge the feasibility and possible limitations of the solution"*

- 强相关：学生必须将柱状图（数学表示） 与企鹅繁殖生态（现实情境） 关联：
  - 理解“平均雏鸟数”反映繁殖成功率；
  - 明白即使该值下降，只要 >0，种群未必灭绝（可能通过其他机制维持）；
  - 认识到“0.4 只/对”意味着部分配对成功，而非全军覆没。
- 这种双向映射（数据 ↔ 生物意义）是第9项的核心。

### 4. 第1项：将数学结果还原到现实世界的情境中

> *"Interpret a mathematical result back into the real world context"*

- 中等相关：例如，读出“2006年平均0.75”后，需理解这意味着“大多数配对成功育雏”，而非仅记住数字；
- 对陈述2的判断尤其依赖此能力：将“平均 < 0.8”转化为“不到80%的配对有雏鸟”——这需要将统计量解释为群体行为比例。

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### 5. 第5项：以多种格式解读数学结果，并根据情境比较或评估不同表示

> *"Interpret mathematical outcomes in a variety of formats... compare or evaluate two or more representations in relation to a situation"*

- 中等相关：虽然仅使用柱状图一种格式，但任务涉及：
  - 跨物种比较（Gentoo vs. Rockhopper vs. Magellanic）；
  - 跨时间比较（2001 vs. 2004）；
  - 数值与阈值比较（>0.6? <0.8?）。
- 这属于在同一图表内进行多维度数据评估，符合“比较多个表示”的广义理解。

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### 6. 第6项：利用现实世界知识判断如何调整或应用数学结果

> *"Use knowledge of how the real world impacts the outcomes... to make contextual judgments about how the results should be adjusted or applied"*

- 弱相关但存在：对陈述3的否定隐含了现实知识：
  - 物种灭绝是复杂过程，不能仅凭繁殖率下降断言；
  - 若学生知道某些企鹅种群具有恢复力，会更坚定判断“False”。
- 虽未要求主动“调整”模型，但拒绝不合理的应用本身是一种情境判断。